

M.Tech. (II Sem.)

1161

CS 503

M.Tech. (SECOND SEMESTER) EXAMINATION, 2025

COMPUTER SCIENCE

CS 503— Advanced Computer Architecture

TIME ALLOWED : THREE HOURS

Maximum Marks— 60

*Attempt six questions in all, selecting two questions from each Section.
All questions carry equal marks.*

SECTION A		
Q. No		Marks
1	Write short notes on following: a) Flynn's taxonomy b) Architectural elements of Computer Design	10
2	Describe the different levels of the memory hierarchy, including registers, cache memory, main memory, and secondary storage. How do they differ in terms of speed, size, and cost?	5
	Compare and Contrast Cache and Cache Coherence	5
3	Explain the role of parallelism in multiprocessor systems. How do techniques like task parallelism and data parallelism improve efficiency?	10
SECTION B		
4	Compare and contrast SISD (Single Instruction, Single Data) and SIMD (Single Instruction, Multiple Data) architectures. In what scenarios is each model used?	10
5	How does symmetric multiprocessing (SMP) differ from asymmetric multiprocessing (AMP)? Provide examples of each	5
	Explain how vector processing works in a multivector system. How does it improve computational performance?	5
6	What factors influence the partitioning of a program in a parallel computing environment? Explain how load balancing is achieved through effective program partitioning. What strategies help in maintaining equal workloads across processors?	10
SECTION C		
7	Discuss the role of vector processing in pipeline architectures. How is it used in high-performance computing?	5

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	Compare and contrast Linear and Non-Linear Pipeline processors with suitable examples.	5
8	What are the different types of pipeline architectures? Explain instruction pipelining, arithmetic pipelining, and superpipelining.	10
9	What are the key differences between pipeline processing and sequential execution? Provide a comparative analysis.	10